

HERMISTON, OR, USA

WMO: 726883

Lat: 45.826N

Lon: 119.261W

Elev: 195

StdP: 99.00

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 04113

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.1	-8.4	-16.0	0.9	-10.1	-12.4	1.3	-7.0	12.3	9.8	11.1	9.1	1.5	160	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	37.3	19.5	35.5	18.7	33.3	18.1	20.6	34.3	19.7	32.8	18.8	31.4	3.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.6	24.2	14.5	10.6	23.7	13.6	10.0	23.5	59.7	34.4	56.7	32.6	53.7	31.3	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.3	8.2	-15.9	40.4	5.3	1.7	-19.7	41.6	-22.8	42.6	-25.7	43.5	-29.6	44.7		
(5)			-16.3	22.6	5.2	2.2	-20.1	24.2	-23.1	25.5	-26.0	26.7	-29.8	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	1.8	3.5	7.6	11.1	15.8	19.7	24.1	23.1	17.9	11.1	5.0	1.1	(6)
(7)		DBStd	8.81	4.75	4.17	3.63	3.06	3.85	3.46	3.24	3.03	3.33	3.83	4.95	5.10	(7)
(8)		HDD10.0	1027	256	185	90	23	2	0	0	0	1	31	162	277	(8)
(9)		HDD18.3	2812	513	416	333	218	99	23	1	3	47	224	401	533	(9)
(10)		CDD10.0	1706	1	2	16	55	181	291	438	406	237	66	11	3	(10)
(11)		CDD18.3	449	0	0	0	1	19	63	181	151	34	1	0	0	(11)
(12)		CDH23.3	6746	0	0	1	36	402	1006	2590	2077	609	25	1	0	(12)
(13)		CDH26.7	3356	0	0	0	4	148	442	1446	1085	228	3	0	0	(13)
(14)	Wind	WSAvg	3.2	2.4	2.8	3.6	3.9	3.8	4.1	3.6	3.4	2.9	2.7	2.6	2.4	(14)
(15)	Precipitation	PrecAvg	233	37	24	22	21	20	13	6	9	12	16	37	34	(15)
(16)		PrecMax	374	244	94	70	55	83	41	22	46	70	44	175	90	(16)
(17)		PrecMin	113	6	0	2	0	0	0	0	0	0	0	2	2	(17)
(18)		PrecStd	54	39	19	14	16	18	10	5	12	14	11	30	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	16.1	22.1	27.1	33.8	37.4	40.4	38.7	34.1	26.3	19.9	16.8	(19)
(20)			MCWB	9.6	10.2	12.5	15.3	17.7	19.8	20.7	19.5	19.0	15.1	13.0	10.7	(20)
(21)		2%	DB	12.3	13.8	18.7	23.8	30.7	34.1	37.8	36.6	31.5	22.8	17.0	13.1	(21)
(22)			MCWB	8.1	8.8	10.9	13.8	16.9	18.6	19.5	19.2	17.8	14.2	11.4	9.2	(22)
(23)		5%	DB	10.2	12.2	16.9	21.2	27.7	31.5	36.2	34.5	29.2	20.6	14.2	10.4	(23)
(24)			MCWB	6.9	7.7	9.9	12.4	15.5	17.3	18.8	18.5	16.7	13.0	9.7	7.3	(24)
(25)		10%	DB	8.2	10.1	14.8	18.7	25.0	28.9	34.0	32.6	27.2	18.4	12.3	8.0	(25)
(26)			MCWB	5.7	6.2	8.9	10.9	14.4	16.3	18.2	17.9	15.8	11.9	8.6	5.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	11.0	13.3	16.1	18.9	21.2	21.9	20.2	17.2	15.0	11.4	(27)	
(28)			MCDWB	13.6	14.9	19.0	24.6	31.6	34.6	37.1	34.6	31.8	23.9	19.0	15.9	(28)
(29)		2%	WB	8.5	9.3	11.7	14.3	17.5	19.4	20.6	20.5	18.7	15.1	11.8	9.4	(29)
(30)			MCDWB	11.5	13.1	17.4	22.4	28.2	32.4	35.2	33.6	29.7	21.3	15.5	13.1	(30)
(31)		5%	WB	7.3	7.9	10.5	12.9	16.3	18.1	19.7	19.5	17.4	13.8	10.4	7.6	(31)
(32)			MCDWB	9.9	11.4	15.7	20.3	26.1	29.7	33.6	32.0	27.7	19.4	13.5	10.0	(32)
(33)		10%	WB	5.9	6.8	9.3	11.5	15.2	16.9	18.8	18.6	16.3	12.5	8.9	5.7	(33)
(34)			MCDWB	8.3	9.7	14.3	17.7	23.2	27.2	32.1	30.7	25.9	17.5	11.9	8.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	10.5	13.0	14.3	15.5	15.7	17.9	17.8	17.1	13.8	9.8	7.1	(35)
(36)			MCDBR	10.1	12.0	15.5	18.9	20.4	20.7	21.3	21.1	20.8	16.4	11.7	10.0	(36)
(37)			MCWBR	6.8	7.7	8.8	9.7	8.4	7.8	7.6	8.1	9.3	9.0	7.2	6.6	(37)
(38)		5% WB	MCDWR	9.6	11.2	14.5	17.6	17.8	19.3	19.6	19.3	18.5	14.6	10.8	9.8	(38)
(39)			MCWBR	6.6	7.4	8.6	9.3	7.5	7.6	7.3	7.7	8.4	8.3	7.0	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.294	0.308	0.341	0.357	0.358	0.358	0.358	0.342	0.315	0.298	0.288	(40)	
(41)		taud	2.552	2.563	2.536	2.417	2.393	2.415	2.419	2.415	2.443	2.526	2.540	2.542	(41)	
(42)		Ebn at Noon	830	900	930	918	910	907	902	890	876	857	811	793	(42)	
(43)		Edh at Noon	63	75	88	109	115	113	112	108	96	77	63	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.25	2.25	3.53	5.20	6.30	7.18	7.71	6.49	4.81	2.97	1.57	1.02	(44)	
(45)		RadStd	0.14	0.28	0.27	0.33	0.46	0.36	0.26	0.36	0.29	0.22	0.13	0.12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	N/A	
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A	

Nomenclature: See separate page